Prosthetic Performance Analysis

INTRODUCTION

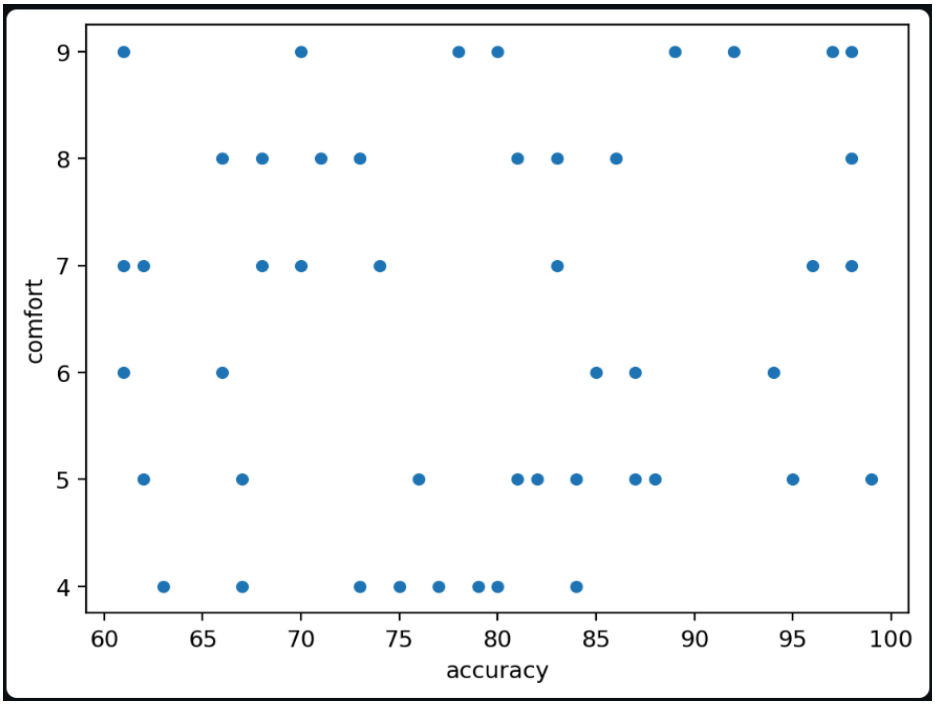
This project analyzes the relationship between prosthetic accuracy and user comfort using a simulated dataset.

TOOLS USED

Python, Pandas, NumPy, Matplotlib,Seaborn,Streamlit

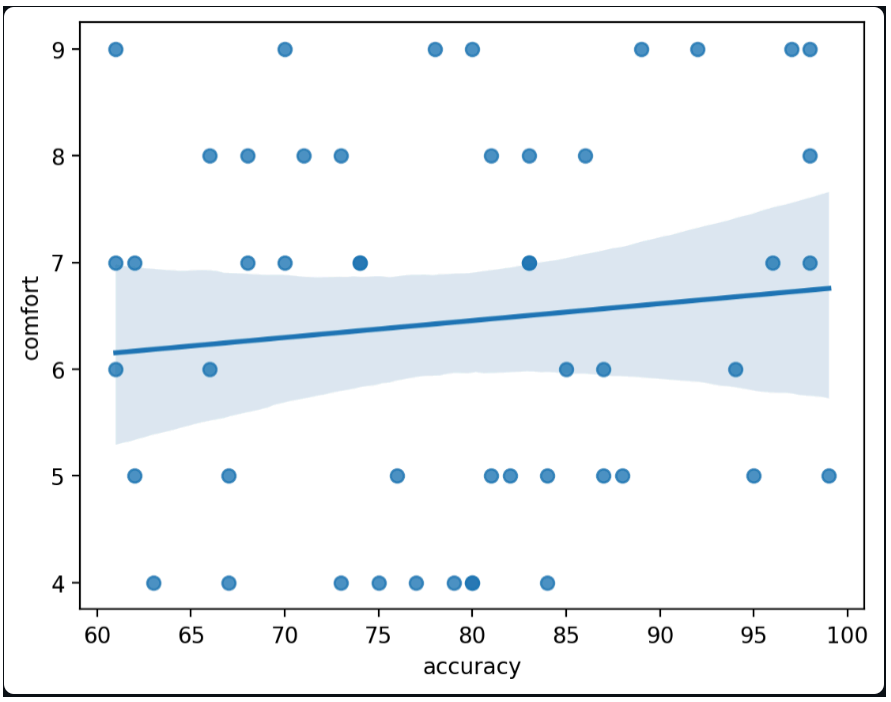
ANALYSIS

1.Scatter Plot



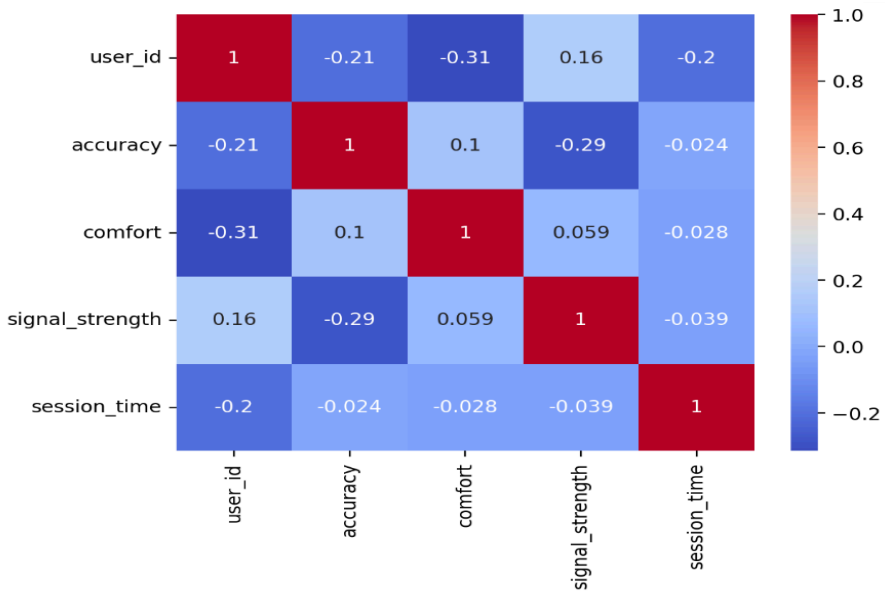
The scatter plot shows no clear pattern between accuracy and comfort.

2. Trend Line



The trend line is nearly flat, indicating a weak relationship.

3. Heatmap



The correlation between accuracy and comfort is around 10%, showing a weak connection.

CONCLUSION

This project analyzed the relationship between prosthetic accuracy and user comfort using data analysis techniques. The results showed a weak correlation of approximately 10%, indicating minimal impact of accuracy on comfort. Visualizations such as scatter plots and regression lines confirmed the absence of a strong relationship. Comfort levels varied even at similar accuracy values, suggesting the influence of other factors. Therefore, improving accuracy alone is not sufficient to enhance overall user comfort.